

SUHAS S

Bengaluru, India | +91 8867406327 | rajsuhas780@gmail.com

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EXPERIENCE

Freo

Jan 2024 – Present

Software Engineer (Backend)

Bengaluru, India

- Eliminated ~₹40L/month in licensing by leading the core LMS migration from proprietary Finflux to Apache Fineract via a decoupled, shadow-mode transformer architecture — zero downtime, backward compatible, and future-portable.
- Designed and delivered end-to-end integrations for 2 lending partners across onboarding, loan creation, repayments and foreclosure – handling each partner’s distinct API contracts and business rules – enabling ₹3Cr+ in monthly disbursements with fault-tolerant, highly available transaction handling.
- Architected an event-driven Video-KYC workflow with Kafka-based async processing, third-party vendor integration, and high-throughput REST APIs; applied the Strategy pattern for partner-specific KYC routing — retaining a key partner disbursing ₹1Cr+/month that mandated VKYC.
- Optimized the high-traffic customer data API by eliminating an N+1 query pattern (replacing per-loan fetches with a single aggregate query), adding Redis caching, parallelizing calls and moving partner-API calls off the request path – cutting latency from ~900 ms to under 100 ms.
- Engineered the Repayments V2 module with idempotent retry and automated settlement reconciliation, reducing repayment-related support tickets by ~75%.
- Integrated the PayU payment gateway as a lower-cost alternative to Razorpay and re-architected the repayment collection flow, reducing per-transaction processing fees (~33% lower on UPI) and cutting monthly payment-processing spend.
- Launched a repayment-time cross-sell module bundling optional add-ons (insurance, credit-report renewals) onto customer EMIs at checkout, using extensible design patterns, contributing ~₹25L/month in incremental revenue.

TECHNICAL SKILLS

Languages: Java, SQL, Python

Frameworks: Spring Boot, Spring Data JPA, Hibernate, Spring AOP, Microservices, REST APIs

Databases: MySQL, PostgreSQL, Redis, DynamoDB, Aurora RDS

Messaging & Streaming: Apache Kafka, AWS SQS

Cloud & DevOps: AWS (ECS, S3, Lambda, API Gateway, CloudWatch, CodeArtifact, ELB), CI/CD, Git, Maven

AI-Assisted Development: GitHub Copilot, Claude, Cursor — prompt engineering, reviewing & hardening AI-generated code

Monitoring: SigNoz, Sentry, Sumo Logic

Concepts: System Design, Distributed Systems, Caching, Event-Driven Architecture, Design Patterns, OOP, SOLID, Concurrency, Scalability, Data Structures & Algorithms

PROJECTS

Field Encryption SDK | *Java 17, Spring Boot, Spring AOP, JPA, JUnit, Maven*

GitHub 2026

- Built a reusable Spring Boot SDK exposing a custom @EncryptDecrypt annotation that, via Spring AOP, transparently encrypts sensitive fields before persistence and decrypts them on retrieval – keeping crypto out of business logic.
- Implemented AES-256-GCM authenticated encryption with a random per-record IV and SHA-256 key derivation, covering round-trip, IV-uniqueness and tamper detection with JUnit tests.

Distributed API Rate Limiter | *Java 17, Spring Boot, Spring AOP, Redis, Lua*

GitHub 2026

- Built a distributed, Redis-backed rate limiter enforcing limits across stateless app servers via atomic Lua scripts that eliminate read-compute-write race conditions under concurrency.
- Implemented pluggable Token Bucket, Sliding Window and Fixed Window algorithms behind a @RateLimit annotation (Spring AOP), with fail-open resilience and standard 429 / Retry-After headers.

EDUCATION

M S Ramaiah University

Dec 2020 – Jul 2024

B.Tech, Computer Science Engineering, CGPA: 9.35/10

Bengaluru, India

CERTIFICATIONS

Programming in Java, Operating System Fundamentals, and Database Management System – NPTEL, Indian Institute of Technology (IIT) Kharagpur